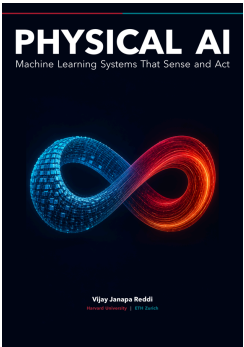


Physical AI Kit

Partner brief for Arduino · successor to the TinyML Kit

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Curriculum physical.mlsysbook.ai · github.com/harvard-edge/physical-ai



Where this sits for you

TinyML Kit — deploy a model on a small board.

Physical AI Kit — stand behind a machine whose learned proposals may **act**, only when independently **permitted**.

UNO Q's dual-brain (Linux MPU + real-time MCU) is the teaching instrument: **MPU proposes · MCU permits**. That matches the silicon—not a bolted-on metaphor.

The ask

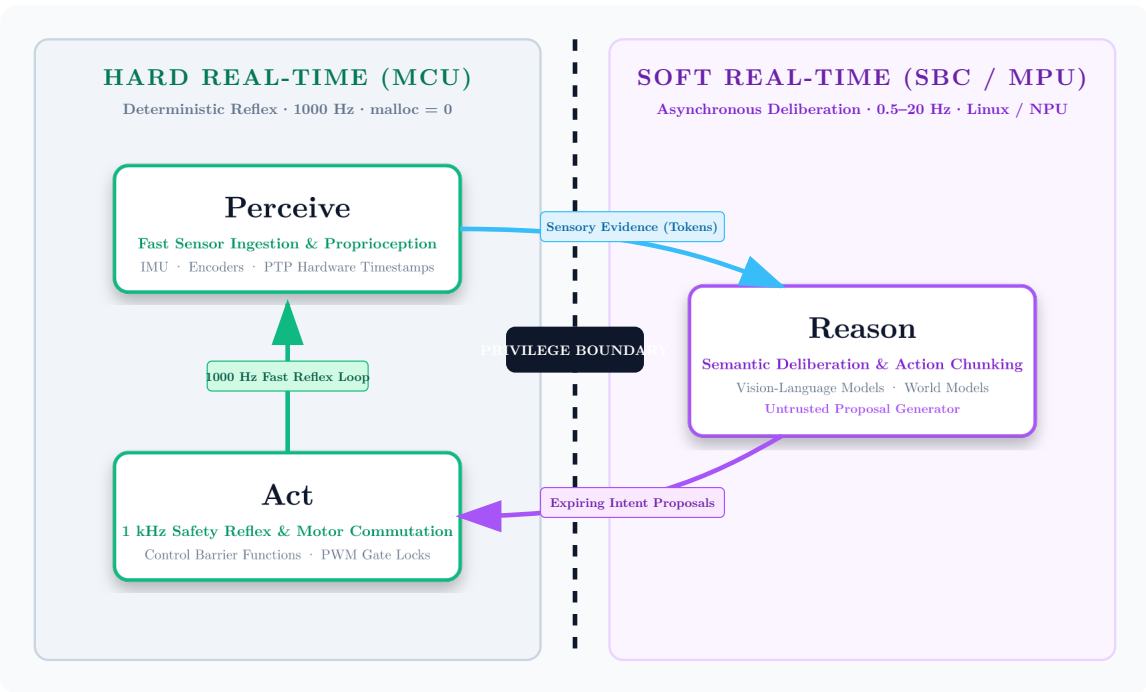
Co-create a **Physical AI Kit** on **Arduino UNO Q**—backed by a university project course and an open book—so the TinyML education line has a clear next chapter. We are **not** asking Arduino to invent pedagogy. We want a **kit that embodies pedagogy that already exists**.

Why now

AI is leaving the screen. Models command motors and contactors. In software, a bad output is a retry; in physics it is momentum —no **ctrl+z**. Embodied demos are everywhere; the missing piece is systems discipline: measure the full path, keep running when intelligence fails, and separate **proposal** from **permission** before kinetic energy hits the world.

What learners practice

Primitive	Question
Loop	Does action change the next observation?
Time	How old may belief be when the move happens?
Budget	What is shared among sensing, inference, link, actuation?
Permission	What may refuse a capable proposal?
Evidence	What measurement supports ship / condition / refuse?



Host proposals vs trusted physical permission

Pedagogy already built

Book	Physical AI: Machine Learning Systems That Sense and Act — 11 chapters + capstone
Course	Project seminar / studio · dossier + midterm + defense · no classical written exam
Labs	Kit contracts: bring-up → measure → MCU enforcer → ship gate
Baseline	Quantize / serve → mlsysbook.ai (not re-taught here)

Signature lab: hang the MPU; the MCU still holds. Capstone ends in **deploy / condition / refuse**.

Kit sketch

Include	Why
Arduino UNO Q (dual-brain)	MPU proposes · MCU permits
Camera + motion / actuation	Real causal loop on the desk
Sensors + safe-idle defaults	Freshness, faults, studio safety
Starter firmware + course link	Failed week ≠ stranded · box → physical.mlsysbook.ai

Box line: PHYSICAL AI KIT · Proposal ≠ permission · Dual-brain UNO Q · physical.mlsysbook.ai

Partnership & next step

Arduino — kit SKU, store, bring-up quality, co-marketing Academic — book + labs + syllabus, proposal–permission
TinyML → Physical AI. method ownership.

1. Kit contents (studio MVP vs retail) · 2. Naming · 3. Pilot → open → store · 4. Narrative: **TinyML deployed the model; Physical AI stands behind the machine.**

Forwardable. Harvard/ETH Physical AI Systems is building the educational successor to TinyML: an open book and project course where students engineer dual-brain agents on Arduino UNO Q—Linux intelligence proposes; the real-time MCU independently permits or refuses action. We want to co-design a **Physical AI Kit** so the same pedagogy can ship worldwide the way the TinyML Kit did—bring-up, measurement, signature enforcer lab, and an evidence-backed deploy / condition / refuse decision.

Next: 30-minute session on kit BOM + out-of-box lab path. · **Email:** vj@eecs.harvard.edu